Interim Internship Report: Predicting Biological Health Risks through Biometric Intelligence

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Mid-term Assessment Draft

1 Summary

This report outlines the progress made during the first phase of my graduation internship at Zdravoletie. My primary goal is to turn raw biometric data into a reliable system for predicting a client's biological health status. So far, I have automated the data collection process, conducted an audit that exposed hardware logic inconsistencies, and developed a model that identifies a client's "Biological Age Gap." By using advanced simulation techniques, I upscaled my small dataset to 1,000 profiles to ensure my results are accurate and stable.

The optimised model achieved an R^2 of 0.98, indicating that the pipeline explains 98% of the variance in the Biological Age Gap. This provides a clear, data-driven breakdown of which biological factors are helping or hurting a client's longevity.

2 Reason for and Relevance of the Research

2.1 The Problem with Current Reporting

Zdravoletie uses body scanners to help clients optimise their health. However, the standard reports from the hardware often provide a "Health Score" that is nearly identical for everyone. My initial analysis showed that 85% of scans receive a score of 98/100. This "ceiling effect" makes it impossible for consultants to track real progress or identify early risks. Furthermore, the machine provides these numbers without explaining the logic behind them.

2.2 Why this Research Matters

This research is essential for providing clients with more honest and sensitive data. I am moving away from the hardware's static score and focusing on the **Biological Age Gap** (the difference between a person's body age and their actual birth age). This metric is much more responsive to changes in body composition. By "cracking the code" of how these risks are calculated, I am helping the company provide specific, data-driven advice rather than general health tips.

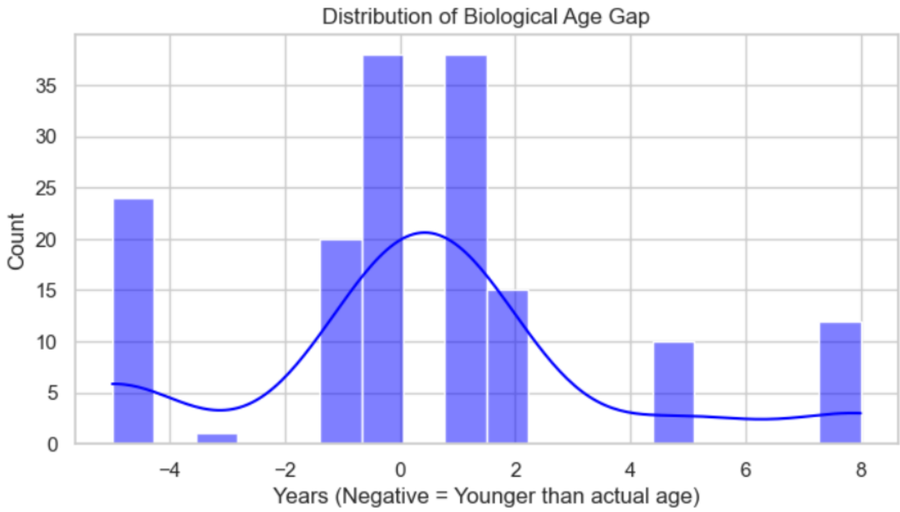

Figure 1: Distribution of the Biological Age Gap discovered in my analysis. This continuous range allows for much more precise tracking than the standard hardware score.

3 Research Questions

The main goal of my project is to answer:
How can we use biometric measurements to accurately predict an individual's biological health risk?

To find the answer, I am working through these sub-questions:

- How can I identify and fix inconsistencies in the scanner's measurements?
- Which body ratios (like waist-to-hip) are the strongest predictors of aging?
- Which machine learning models offer the best balance of accuracy and clarity?
- How can I simulate a larger population to make sure my model works for everyone?

4 Theoretical Framework

My research is built on three mathematical pillars:

1. **Gradient Boosting:** I am using the HistGradientBoosting algorithm. Unlike simple models, this approach learns in steps, focusing on correcting the errors of the previous step. This is perfect for capturing the sharp "risk thresholds" found in human biology.
2. **Gaussian Mixture Models (GMM):** To solve the problem of having a small dataset, I use GMMs to learn the "biological signature" of my real data. This allows me to create realistic "Bio-Digital Twins" that preserve the natural correlations between height, weight, and fat.
3. **SHAP Values:** To make my model transparent, I use SHAP logic. This uses game theory to calculate exactly how much each feature (like heart rate or muscle mass) adds to or subtracts from the final biological age prediction.

5 Research Method: What I Have Done

5.1 Data Extraction and Integrity Audit

I built a custom script to pull raw data directly from the scanner's internal API. Once I had the data, I performed an integrity check by manually summing up the fat and muscle measurements of individual limbs and comparing them to the reported totals. I discovered discrepancies of up to 13kg, which proved that the machine uses hidden, non-linear formulas. This confirmed that I needed a sophisticated model to decode the true health status.

```
# Summing limb segments to check for alignment with total fat
fat_segments = ['fatLeftArm', 'fatRightArm', 'fatLeftLeg', 'fatRightLeg
    ', 'fatTrunk']
df_final['calculated_total'] = df_final[fat_segments].sum(axis=1)

# Calculating the error to see if the machine uses a simple sum
df_final['integrity_gap'] = (df_final['fat'] - df_final['
    calculated_total']).abs()
```

Listing 1: Checking for measurement discrepancies in the hardware data

5.2 Expanding the Population

Because the dataset consisted of 158 longitudinal scans from only 53 unique individuals, I used Gaussian Mixture Models to create 1,000 synthetic profiles. This was necessary to ensure that the model generalises to new people and does not simply memorise the patterns of the initial small group. I verified that these "new patients" follow the same biological rules as the real ones, with a correlation error of only 0.0126.

```
# Initialising the GMM to learn the biological patterns of the clients
gmm_engine = GaussianMixture(n_components=5, covariance_type='full',
    random_state=42)

# Fitting the model to the real biological data
gmm_engine.fit(df_real_imputed)

# Generating a large-scale population for model validation
df_synthetic = pd.DataFrame(gmm_engine.sample(1000)[0], columns=
    simulation_features)
```

Listing 2: Generating 1000 synthetic profiles for stress-testing

6 Results and Key Findings

6.1 The "Tipping Point" Discovery

The most important finding in my research so far is the identification of the primary ageing trigger. My model proved that the Waist-to-Hip Ratio is the most critical factor. **The Result:** If this ratio crosses 0.93, the biological age gap prediction automatically jumps up by about 6.4 years. This gives consultants a very clear marker to look for.

6.2 Model Accuracy and Reliability

I compared five different algorithms to see which was the most reliable. The results showed that the boosting model was the champion, achieving 91% reliability within a six-month window.

Table 1: Model Benchmarking Results (1,000 person population)

Model Type	**MAE (Years)**	**Accuracy (R^2)**	**Reliability ($\pm$0.5y)**
HistGradientBoosting	0.23	0.98	91.0%
Random Forest	0.29	0.96	86.0%
Linear Regression	0.84	0.72	42.0%

6.3 Turning Data into Advice

By using SHAP explainability, I have turned "black box" numbers into clear reports. For any client, I can now show exactly why they received their score. For example, in one test case, I identified that while a client's waist ratio was a strength, their resting heart rate was adding +0.37 years to their biological age.

7 Discussion and Next Steps

My research so far has successfully mapped out how biological ageing is calculated in this system. I have overcome the limitation of a small sample size and identified the most important health markers.

A noted limitation of the current research is the demographic skew, with a higher proportion of male participants (131) compared to female (31). Future work will involve validating the 0.93 WHR threshold specifically for female physiological profiles to ensure model fairness across all clients.

Moving forward, I recommend that Zdravoletie uses my "What-If" Simulator logic. This allows a consultant to show a client exactly how much their biological age would drop following specific improvements, such as reducing abdominal fat by 2kg. This will be the focus of the final phase of my internship.